

Jacobian Descent For Multi-Objective Optimization

Valérian Rey — Simplex Lab
valerian.rey@gmail.com
Joint work with Pierre Quinton

February 10, 2026



github.com/TorchJD/torchjd



torchjd.org



arxiv.org/pdf/2406.16232

Table of contents

Theory

- Multi-objective optimization

- Pareto optimality

- First-order Taylor approximation

- Jacobian descent

- Unconflicting projection of gradients (UPGrad)

- Formal definition of UPGrad

Applications and experimentation

- Optimization trajectories

- Multi-task learning

- Experimental results: per-sample JD

- Future directions

Theory

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

$$\begin{aligned} \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow \quad u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

$$\begin{aligned} \mathbf{u} &\preceq \mathbf{v} \\ \Leftrightarrow \quad u_i &\leq v_i \quad \forall i \in [m] \end{aligned}$$

Examples:

$$\blacktriangleright \begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

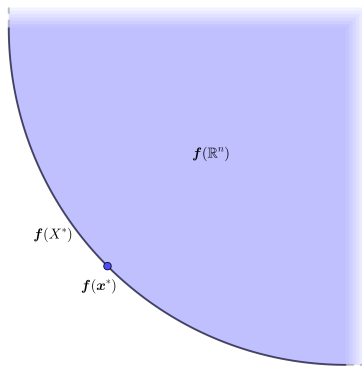
$$\begin{aligned} \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow \quad u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

Examples:

▶ $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

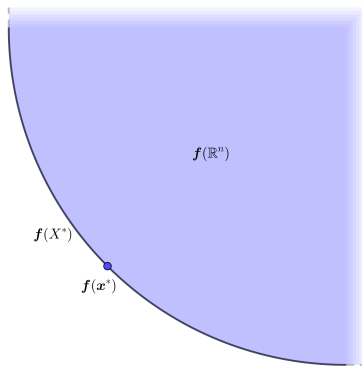
▶ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



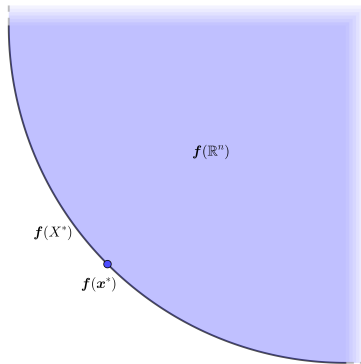
Pareto optimality

Pareto set: X^*



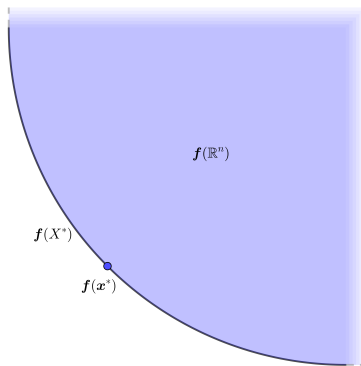
Pareto optimality

Pareto set: X^*
Pareto Front: $f(X^*)$



Pareto optimality

Pareto set: X^*
Pareto Front: $f(X^*)$



$$x^* \in X^* \quad \Leftrightarrow \quad \forall y \in \mathbb{R}^n, f(y) \preceq f(x^*) \rightarrow f(y) = f(x^*)$$

First-order Taylor approximation

For any $\mathbf{x} \in \mathbb{R}^n$, the Jacobian of $\mathbf{f}$ at $\mathbf{x}$, is

$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

First-order Taylor approximation

For any $\mathbf{x} \in \mathbb{R}^n$, the Jacobian of $\mathbf{f}$ at $\mathbf{x}$, is

$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

For a small enough update $\mathbf{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\mathbf{f}(\mathbf{x} + \mathbf{y}) \approx \mathbf{f}(\mathbf{x}) + \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathbf{y}$$

First-order Taylor approximation

For any $\mathbf{x} \in \mathbb{R}^n$, the Jacobian of $\mathbf{f}$ at $\mathbf{x}$, is

$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

For a small enough update $\mathbf{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\mathbf{f}(\mathbf{x} + \mathbf{y}) \approx \mathbf{f}(\mathbf{x}) + \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathbf{y}$$

We select $\mathbf{y} = -\eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

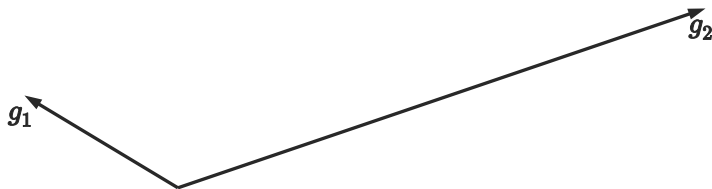
Input: $\mathbf{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$

for $t \leftarrow 1$ **to** T **do**

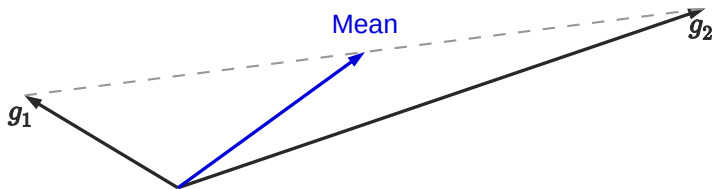
$\mathbf{x} \leftarrow \mathbf{x} - \eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x}))$

Output: $\mathbf{x}$

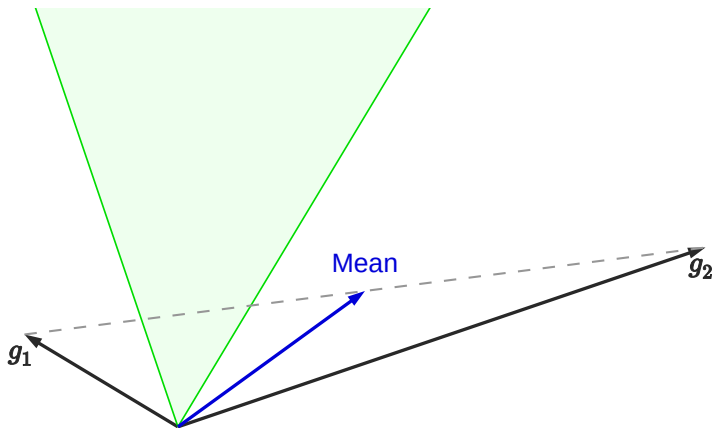
Unconflicting projection of gradients (UPGrad)



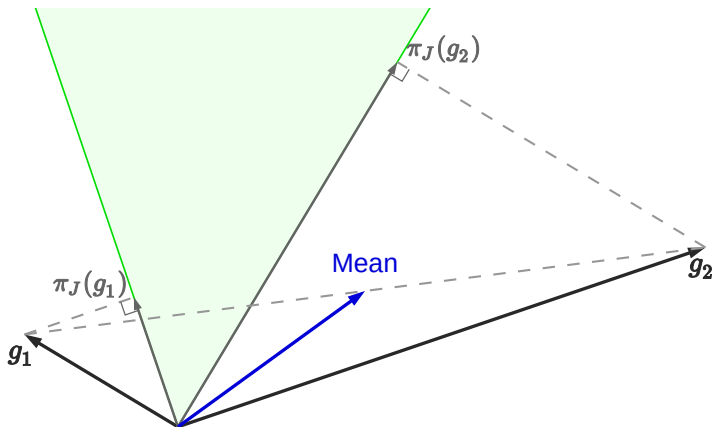
Unconflicting projection of gradients (UPGrad)



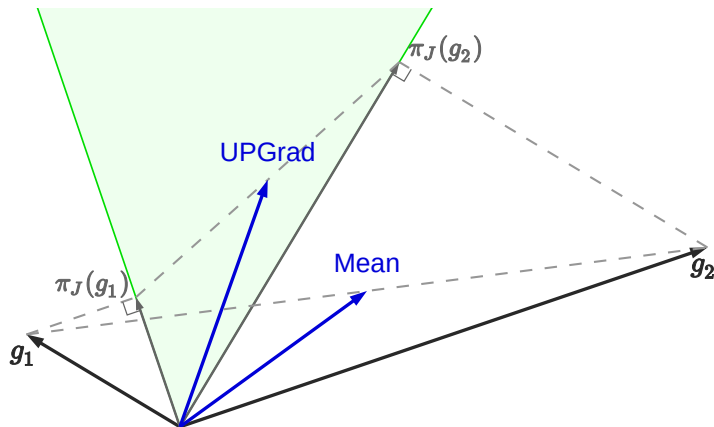
Unconflicting projection of gradients (UPGrad)



Unconflicting projection of gradients (UPGrad)



Unconflicting projection of gradients (UPGrad)



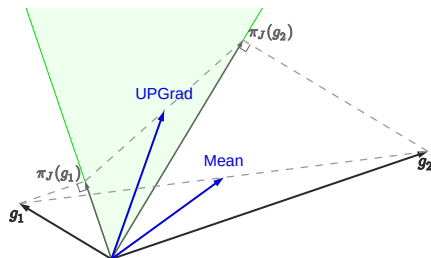
Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\mathbf{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\mathbf{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\mathbf{x} \in \mathbb{R}^n$, the projection of $\mathbf{x}$ onto C_J is

$$\pi_J(\mathbf{x}) = \arg \min_{\mathbf{y} \in C_J} \|\mathbf{x} - \mathbf{y}\|^2$$

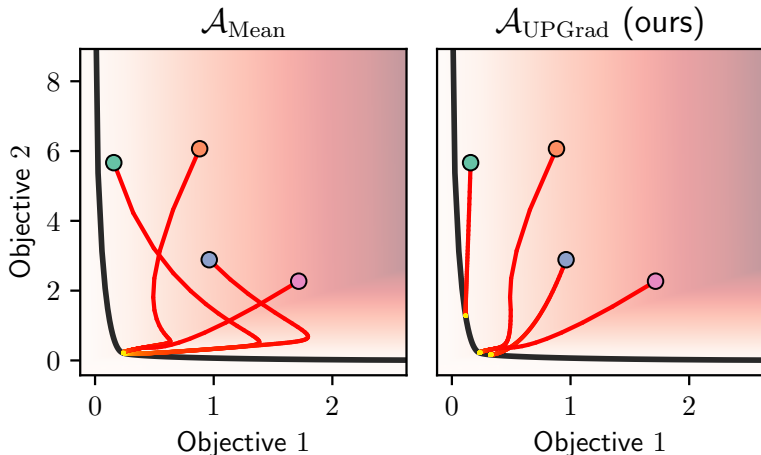
$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(J^\top \mathbf{e}_i)$$



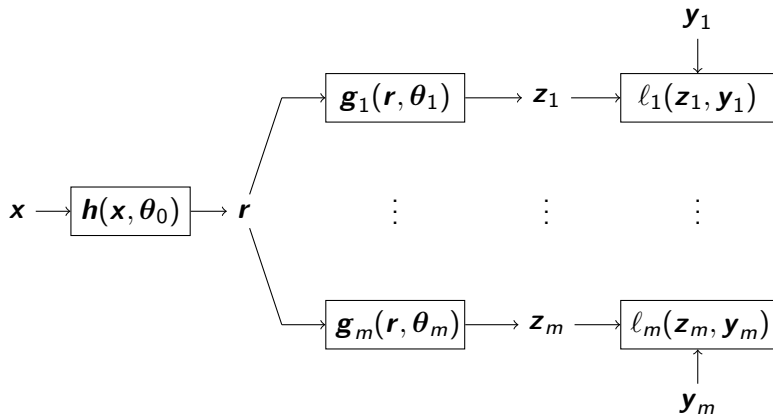
Applications and experimentation

Optimization trajectories

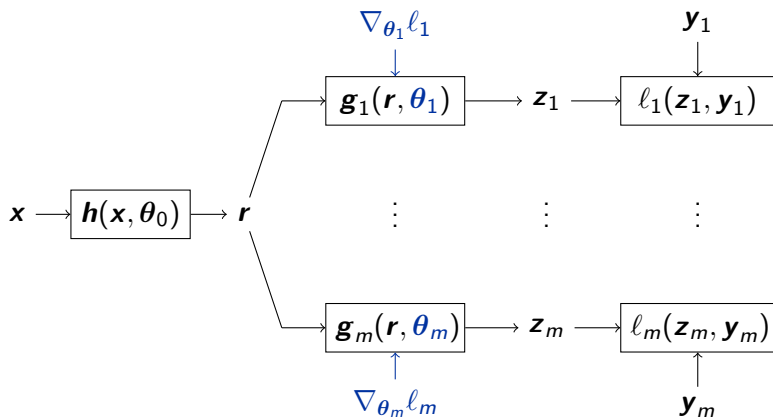
Some convex quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with conflicting gradients.



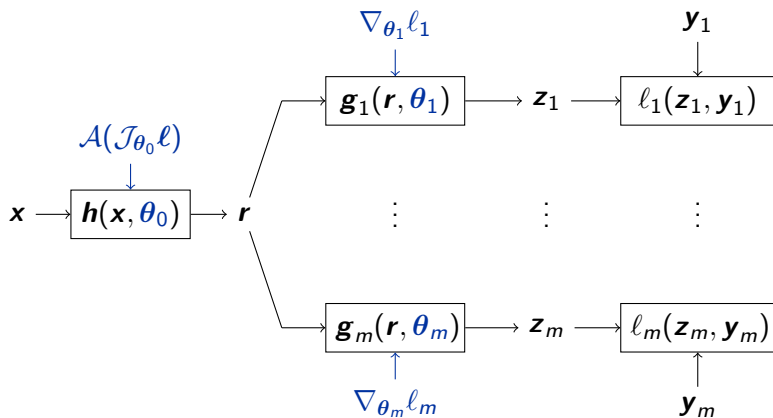
Multi-task learning



Multi-task learning



Multi-task learning



Multi-task learning: code

```
backbone = Linear(10, 5)
head1 = Linear(5, 1)
head2 = Linear(5, 1)

for x, y1, y2 in dataloader:
    features = backbone(x)
    y_pred1 = head1(features)
    y_pred2 = head2(features)
    loss1 = loss_fn(y_pred1, y1)
    loss2 = loss_fn(y_pred2, y2)

    mtl_backward([loss1, loss2], features)
    jac_to_grad(backbone.parameters(), UPGrad())
    optimizer.step()
    optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/mtl.

Experimental results: per-sample JD

Each element of the batch has its own loss.

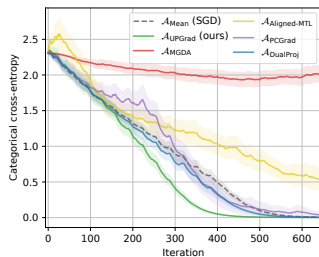
Experimental results: per-sample JD

Each element of the batch has its own loss.

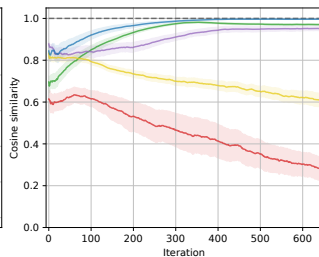
CIFAR-10



Training Loss



Update Similarity



Future directions

Future directions

Implementation

- ▶ Make the `autogram` engine faster.

Implementation

- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.

Implementation

- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.
- ▶ Improve the interface of TorchJD.

Implementation

- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.
- ▶ Improve the interface of TorchJD.

Applications

- ▶ Are there multi-task learning problems with actual conflict?

Implementation

- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.
- ▶ Improve the interface of TorchJD.

Applications

- ▶ Are there multi-task learning problems with actual conflict?
- ▶ Could JD (per-sample) help training diffusion models?

Implementation

- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.
- ▶ Improve the interface of TorchJD.

Applications

- ▶ Are there multi-task learning problems with actual conflict?
- ▶ Could JD (per-sample) help training diffusion models?
- ▶ Could JD (per-depth) help training recurrent models?

Implementation

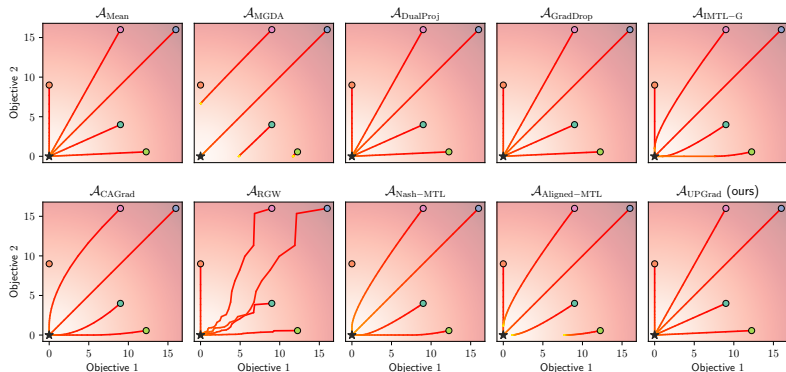
- ▶ Make the `autogram` engine faster.
- ▶ Make a better implementation of $\mathcal{A}_{\text{UPGrad}}$.
- ▶ Improve the interface of TorchJD.

Applications

- ▶ Are there multi-task learning problems with actual conflict?
- ▶ Could JD (per-sample) help training diffusion models?
- ▶ Could JD (per-depth) help training recurrent models?
- ▶ Find new applications.

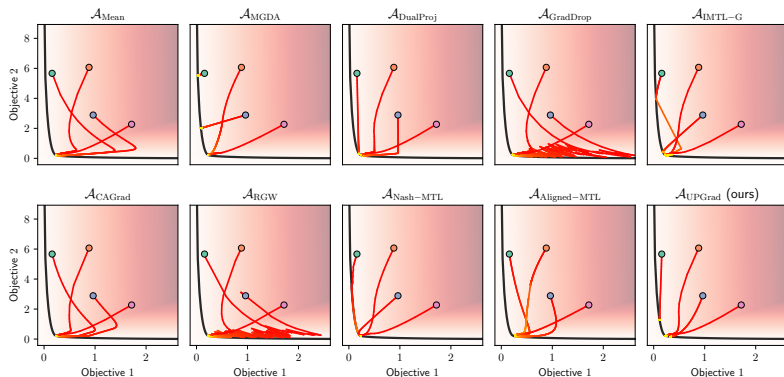
Bonus: Element-wise quadratic function trajectories

Minimize $\mathbf{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Bonus: Convex quadratic form trajectories

Some convex quadratic function $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with conflicting gradients.



Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

The weights only depend on the (small) Gramian
 $\mathcal{G}\mathbf{f}(\mathbf{x}) = \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathcal{J}\mathbf{f}(\mathbf{x})^\top$.

Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

The weights only depend on the (small) Gramian
 $\mathcal{G}\mathbf{f}(\mathbf{x}) = \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathcal{J}\mathbf{f}(\mathbf{x})^\top$.

We can compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

The weights only depend on the (small) Gramian
 $\mathcal{G}\mathbf{f}(\mathbf{x}) = \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathcal{J}\mathbf{f}(\mathbf{x})^\top$.

We can compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the `autogram` engine to compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ efficiently.

Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

The weights only depend on the (small) Gramian
 $\mathcal{G}\mathbf{f}(\mathbf{x}) = \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathcal{J}\mathbf{f}(\mathbf{x})^\top$.

We can compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the `autogram` engine to compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ efficiently.

This saves a lot of memory, so it's often much faster.

Bonus: Gramian-based JD

$\mathcal{A}_{\text{UPGrad}}(\mathcal{J}\mathbf{f}(\mathbf{x}))$ is a linear combination of the rows of $\mathcal{J}\mathbf{f}(\mathbf{x})$.

The weights only depend on the (small) Gramian
 $\mathcal{G}\mathbf{f}(\mathbf{x}) = \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathcal{J}\mathbf{f}(\mathbf{x})^\top$.

We can compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ directly, extract the weights, combine the losses, and do a step of gradient descent.

We have developed the `autogram` engine to compute $\mathcal{G}\mathbf{f}(\mathbf{x})$ efficiently.

This saves a lot of memory, so it's often much faster.

Applicable to most other aggregators from the literature.